

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

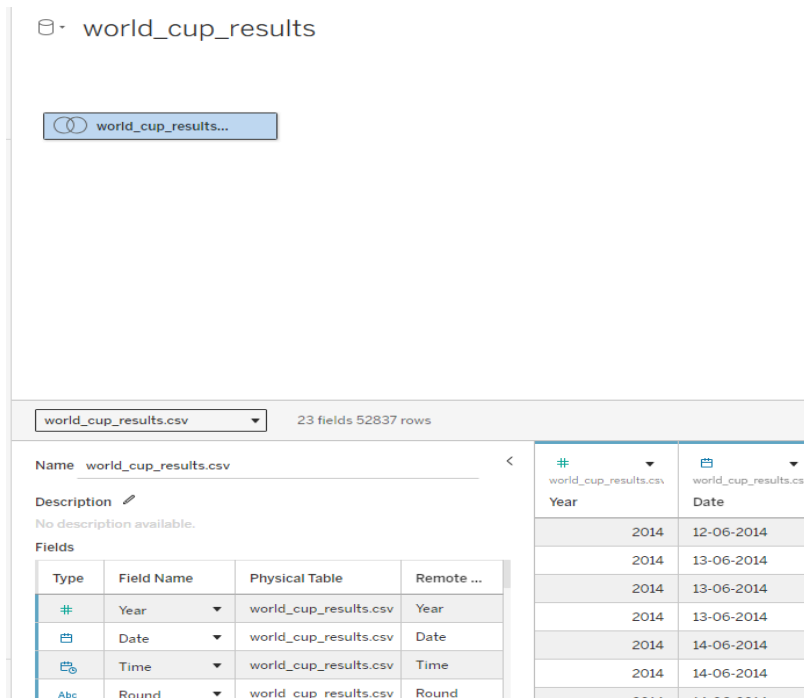
Practical – II

Roll No.: 52	Name: Divyanjali yadav
Class: TYIT	Batch:
Date of Assignment: 31-07-2026	Date/Time of Submission: 31-07-2026

Working with Data Sources and Data Preparation in Tableau

a) Connect to different data sources (Excel, CSV, Database)

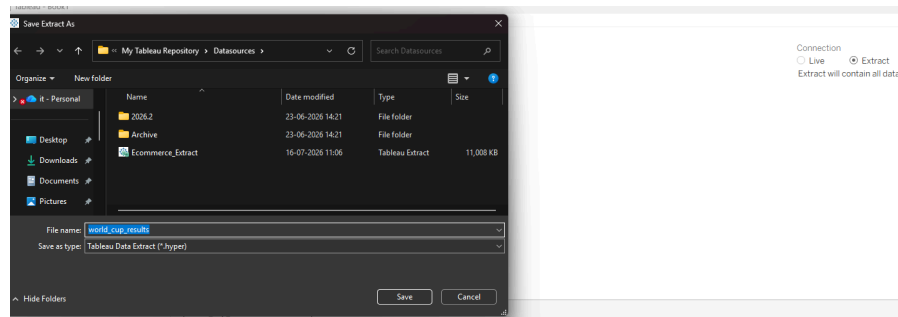
1. Browse and select the file (e.g., **world_cup_results.csv**).
2. The data source loads into the canvas showing the table icon, field list, and a preview grid (as seen in your screenshot — 23 fields, 52,837 rows).



Insight: Tableau can pull data from flat files, spreadsheets, and databases into one interface, letting you standardize analysis regardless of source format.

b) Create and use data extracts vs live connections

1. On the data source screen, toggle **Connection: Live** or **Extract** (top-right).
2. Choosing **Extract** compresses data into Tableau's optimized **.hyper** format.
3. Go to **Data** → **Extract** → **Compute/Save As**, name the file, and save (as shown: "Save Extract As" dialog).



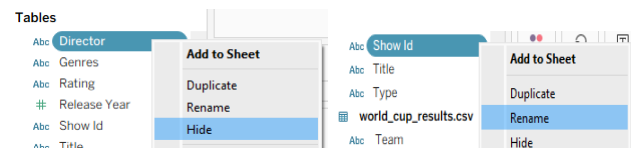
Insight:

- **Live connection** = always queries the original source directly → real-time data, but slower on large datasets.
- **Extract** = a snapshot stored locally → much faster performance, works offline, but needs manual/scheduled refresh to stay current.

c) Manage metadata (rename, hide, change data types)

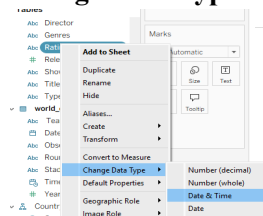
1. **Rename** → select "Rename," type new name (e.g., renaming "Director").
2. **Hide** → select "Hide" to remove unused fields ("Show Id") from view without deleting them.
3. **Change Data Type** → hover over the data-type icon next to a field → choose Number, Date, String, etc. (e.g., "Rating" changed to Date & Time).

Rename



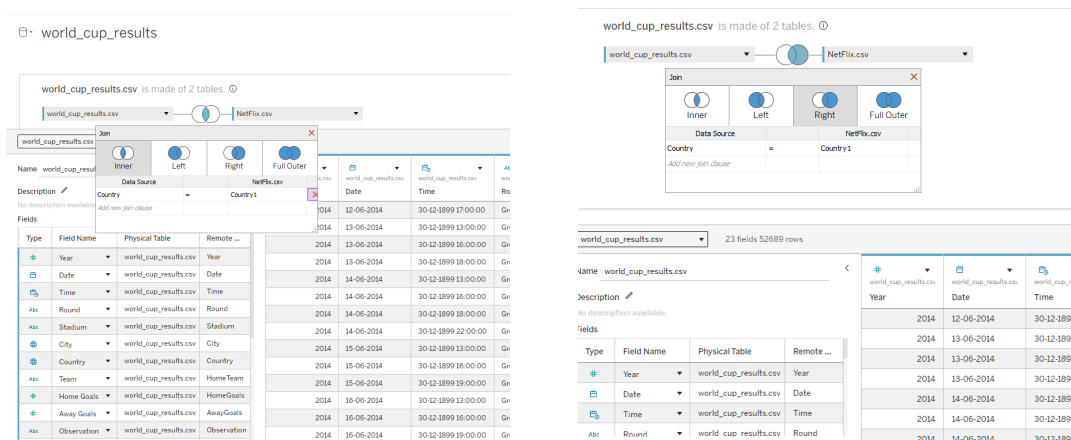
Hide

Change Data Type



Insight: Cleaning metadata early prevents errors in calculations (e.g., a numeric field stored as text won't aggregate) and keeps the field list uncluttered for end users.

d) Perform joins (inner, left, right) on multiple tables



world_cup_results.csv is made of 2 tables. ⓘ

world_cup_results.csv

Join

Inner Left Right Full Outer

Data Source = NetFlx.csv

Country = Country1

Add new join clause

world_cup_results.csv 23 fields 52837 rows

Name world_cup_results.csv

Description No description available.

Fields

Type	Field Name	Physical Table	Remote ...
#	Year	world_cup_results.csv	Year
📅	Date	world_cup_results.csv	Date
🕒	Time	world_cup_results.csv	Time
🏟️	Round	world_cup_results.csv	Round
🏟️	Stadium	world_cup_results.csv	Stadium
🌐	City	world_cup_results.csv	City
🌐	Country	world_cup_results.csv	Country
👤	Team	world_cup_results.csv	Home Team
👤	Home Goals	world_cup_results.csv	Home Goals
👤	Away Goals	world_cup_results.csv	Away Goals
👤	Observation	world_cup_results.csv	Observation

Year	Date	Time	Round	Group
2014	12-06-2014	30-12-1899 17:00:00	Group A	Group A
2014	13-06-2014	30-12-1899 13:00:00	Group A	Group A
2014	13-06-2014	30-12-1899 16:00:00	Group B	Group B
2014	13-06-2014	30-12-1899 18:00:00	Group B	Group B
2014	14-06-2014	30-12-1899 13:00:00	Group C	Group C
2014	14-06-2014	30-12-1899 16:00:00	Group D	Group D
2014	14-06-2014	30-12-1899 18:00:00	Group D	Group D
2014	14-06-2014	30-12-1899 22:00:00	Group C	Group C
2014	15-06-2014	30-12-1899 13:00:00	Group E	Group E
2014	15-06-2014	30-12-1899 16:00:00	Group E	Group E
2014	15-06-2014	30-12-1899 19:00:00	Group F	Group F
2014	16-06-2014	30-12-1899 13:00:00	Group G	Group G
2014	16-06-2014	30-12-1899 16:00:00	Group F	Group F
2014	16-06-2014	30-12-1899 19:00:00	Group G	Group G

Insight:

- **Inner** → only matching rows from both tables.
 - **Left** → all rows from the primary table + matches from the second.
 - **Right** → all rows from the secondary table + matches from the first.
 - **Full Outer** → all rows from both, matched where possible.
- Joins physically merge tables at the row level — used when data shares a common key and granularity.

e) Apply data blending between two sources

Bring in a **primary data source** (e.g., **world_cup_results**) and a **secondary source** (e.g., **Iris**) — each as separate connections (not joined).

On a worksheet, use fields from both sources.

Tableau shows an orange checkmark/link icon on the shared field (**Country**) — go to **Data → Edit Blend Relationships** to set/customize the linking field.

Tableau - Book1

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Data Analytics

world_cup_results

Search

Tables

- Director
- Genres
- Rating
- Release Year
- Show Id
- Title
- Type
- world_cup_results.csv
 - Team
 - Date
 - Observation
 - Round
 - Stadium
 - Time
 - Year
 - Country, City
 - Country

Filters

Marks

Automatic

Color Size Text

Detail Tooltip

Columns

Rows

Round

Sheet 1

Final

First round

Group 1

Group 2

Group 3

Group 4

Group 5

Group 6

Group A

Group B

Group C

Group D

Group E

Group F

Group G

Group H

Match for third place

Blend Relationships

Blend relationships determine how data from secondary data sources are joined with primary data sources.

Primary data source: world_cup_results

Secondary data source: Iris

Automatic Custom

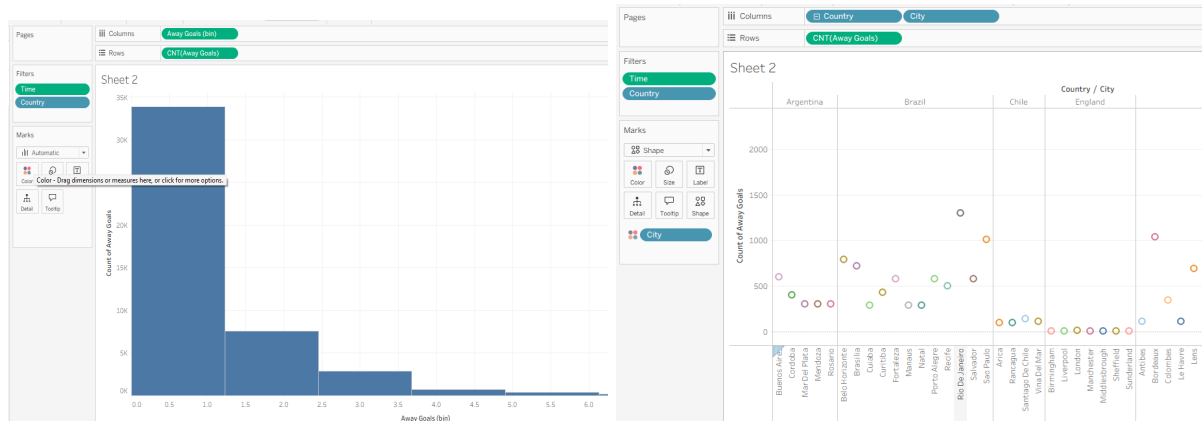
Country Id

Add... Edit... Remove

OK Cancel

f) Implement different types of filters (dimension, measure, date)

1. Drag a field to the **Filters** shelf.
2. **Dimension filter** (e.g., Country/City) → select specific categorical values to include/exclude.
3. **Measure filter** (e.g., Away Goals) → set a range, min/max, or aggregation condition.
4. **Date filter** (e.g., Time) → choose relative date, range of dates, or specific date parts (year/quarter)



g) Optimize performance using extracts and filtering

1. Convert the live connection to an **Extract** (reduces load per query since data is pre-aggregated/compressed).
2. Apply **extract filters** (Data Source page → Filters → Edit/Add) to limit rows pulled in, e.g., filtering down to 288 rows from 52,837.
3. Save/refresh the extract so the workbook only loads the filtered subset.

world_cup_results

Connection
☒ Live
☐ Extract

Filters
 1 | Edit

world_cup_results...

world_cup_results.csv 24 fields 288 rows

100 rows

#	Year	Date	Time	Round	Stadium	City	Country
1	2014	16-06-2014	30-12-1899 16:00:00	Group F	Arena da Baixada	Curitiba	Brazil
2	2014	20-06-2014	30-12-1899 19:00:00	Group E	Arena da Baixada	Curitiba	Brazil
3	2014	23-06-2014	30-12-1899 13:00:00	Group B	Arena da Baixada	Curitiba	Brazil
4	2014	26-06-2014	30-12-1899 17:00:00	Group H	Arena da Baixada	Curitiba	Brazil
5	2014	16-06-2014	30-12-1899 16:00:00	Group F	Arena da Baixada	Curitiba	Brazil
6	2014	20-06-2014	30-12-1899 19:00:00	Group E	Arena da Baixada	Curitiba	Brazil
7	2014	23-06-2014	30-12-1899 13:00:00	Group B	Arena da Baixada	Curitiba	Brazil